

To:

The Editors-in-Chief
Computer Methods in Applied Mechanics and Engineering (CMAME)
 Elsevier Science

Date: September 25, 2026

Subject: Submission of Original Research Article

Dear Editors-in-Chief,

We are pleased to submit our original research manuscript entitled:

**“Stateful Selective Maintenance of Two-Level Schwarz Preconditioners
 for Evolving Sparse Linear Systems”**

for consideration for publication as a regular research article in *Computer Methods in Applied Mechanics and Engineering* (CMAME).

Context and Motivation:

In transient continuum mechanics, thermal transport, and multiphysics simulations, numerical time integration generates sequences of large, sparse symmetric positive definite linear algebraic systems $A_t x_t = b_t$. In problems characterized by moving phase-change fronts, localized thermal softening, non-Newtonian boundary layer evolution, or progressive damage, the physical differential operator evolves intensely across a localized sub-region while the vast background remains quasi-static. Traditional parallel domain decomposition methods face an expensive operational dilemma: recomputing all subdomain sparse direct factorizations at every time step prevents Krylov iteration inflation but incurs heavy setup overhead (often 25%–45% of total execution time in three dimensions); conversely, freezing the initial preconditioner statically costs zero setup but triggers severe spectral mismatch and solver divergence.

Core Scientific Contributions:

In this work, we introduce a stateful selective maintenance framework for Two-Level Overlapping Symmetric Weighted Additive Schwarz (S-AS) preconditioners accelerated by sparse direct subdomain solvers (PETSc/MUMPS) and exact coarse Galerkin projections. The key contributions include:

1. **Persistent Stateful Architecture & Conditional Spectral Stability:** Rather than rebuilding from scratch or freezing statically, the preconditioner is maintained as a persistent stateful object. We establish a rigorous conditional perturbation theorem (Proposition 1) proving that bounded relative drift on skipped subdomains ($\|E_j\|_2 \leq \varepsilon < 1/2$) strictly bounds the effective condition number by $\kappa(M_t^{-1} A_t) \leq \frac{1}{1-2\varepsilon} \kappa(M_{\text{exact}}^{-1} A_t)$.
2. **Inexpensive Online Monitoring:** An $\mathcal{O}(n_i)$ diagonal operator-drift proxy senses local perturbations with negligible computational overhead ($< 0.026\%$ of total runtime, $< 0.28\%$ of saved setup time), showing strong empirical agreement in subdomain ranking (Pearson $r = 1.0000$, Spearman $\rho_s \in [0.79, 0.98]$ on tested benchmarks).
3. **Mass- α Cumulative-Drift Policy & Coarse Synchronization:** Disturbed subdomains are selected via a cumulative staleness metric, while coarse-space Galerkin synchronization preserves low-frequency error damping. Multi-step ablation demonstrates coarse synchronization is essential to prevent long-wave degradation.
4. **Cross-Operator Generalization:** We formalize the extension to vector continuum mechanics via 3-D damaged linear elasticity, establishing that the stateful maintenance frame-

work is an algebraic operator strategy governed by the spatial sparsity of time-derivative support.

5. **Subdomain Granularity Resolution Effect:** A systematic study across $N_{\text{sub}} \in \{8, 27, 64\}$ quantifies how spatial decomposition resolution enables sharper front isolation, reducing the refactored ratio to 31.8% and eliminating Krylov iteration penalty (47.0 vs. 47.0 iterations).
6. **Certified 3-D Flagship Speedup:** On a flagship 3-D benchmark with 110,592 degrees of freedom (48^3), a 63.9% reduction in preconditioner setup translates into up to a 21.8% reduction in end-to-end simulation runtime ($1.24\times \sim 1.28\times$ net speedup), certified under an independent algebraic residual bound of $\text{RelRes} \leq 3.89 \times 10^{-10} \ll 1.0 \times 10^{-8}$.

Delineation from Contemporary Literature:

We clearly delineate our work from contemporary Krylov subspace recycling (e.g., Hanek, Papež & Šístek, *CMAME* 452 (2026), 118788). While Krylov recycling deflates an invariant operator $Ax_t = b_t$, our framework actively repairs the preconditioner to track a time-evolving operator $A_t x_t = b_t$ ($A_t \neq A_{t-1}$).

Reproducibility Statement:

In accordance with CMAME’s open science and reproducibility standards, the complete source code, benchmark drivers, and verification suites are made publicly available under the MIT license at <https://github.com/Deepsleepinger/JSR> under the official release tag `v1.2-cmame-final`.

We confirm that this manuscript represents original work, has not been published previously, and is not currently under consideration for publication elsewhere. All authors have approved the submission.

Sincerely,

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